

WENZER QIN

CONTACT INFORMATION

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EMPLOYMENT

2024-present Simons Society of Fellows, New York, NY
New York University

EDUCATION

2019-2024 Massachusetts Institute of Technology, Cambridge, MA
Advisor: Prof. Tracy SLATYER
Ph.D., Physics; GPA: 5.0/5.0
2015-2019 Johns Hopkins University, Baltimore, MD
Advisors: Prof. Marc KAMIONKOWSKI, Prof. Nadia ZAKAMSKA, Prof. Andrei GRITSAN
B.S., Physics; B.A., Mathematics; GPA: 3.99/4.0 (Major GPA: 4.0/4.0)

AWARDS AND HONORS

- 2023 · Simons Society of Fellows and Flatiron Research Fellowship
- 2021 · MIT Physics Graduate Student Service Award
- 2019 · NSF Graduate Research Fellowship
 - LeRoy Apker Award Finalist
 - C. M. Clay and Henry W. Kendall Fellowships (MIT First Year Fellowships)
 - Donald E. Kerr Memorial Award for outstanding graduating senior in Physics
 - J.J. Sylvester Award for outstanding graduating senior in Mathematics
- 2018 · Barry Goldwater Scholarship
 - Caltech's FUTURE of Physics program for outstanding undergraduate women in physics
 - Johns Hopkins STAR Award for summer research
- 2017 · Inducted into Sigma Pi Sigma ($\Sigma \Pi \Sigma$)
 - Maryland Space Grant recipient for summer research
 - 2nd Place Presentation, Maryland Space Grant Consortium Student Research Symposium

PUBLICATIONS (* INDICATES ALPHABETICAL AUTHORSHIP ORDER)

- [15] **W. Qin**, K.-F. Chen, K. Schutz, and A. Liu. "Effective bias expansion for circumventing 21 cm foregrounds". 8 2025. [arXiv:2508.13268 \[astro-ph.CO\]](https://arxiv.org/abs/2508.13268)
- [14] **W. Qin**, S. Kumar, P. Natarajan, and N. Weiner. "Not-quite-primordial black holes". June 2025. [arXiv:2506.13858 \[astro-ph.CO\]](https://arxiv.org/abs/2506.13858)
- [13] **W. Qin**, K. Schutz, O. Rosenstein, S. O'Neil, and M. Vogelsberger. "Supersizing hydrodynamical simulations of reionization using perturbative techniques". Apr 2025. [arXiv:2504.02929 \[astro-ph.CO\]](https://arxiv.org/abs/2504.02929)
- [12] C. Xu, **W. Qin**, and T. R. Slatyer. "CMB limits on decaying dark matter beyond the ionization threshold". *Phys. Rev. D*, 110(12):123529, 2024. [arXiv:2408.13305 \[astro-ph.CO\]](https://arxiv.org/abs/2408.13305)

- [11] **W. Qin**, J. B. Munoz, H Liu, and T. R. Slatyer. “Birth of the first stars amidst decaying and annihilating dark matter”. *Phys. Rev. D*, 109(10):103026, 2024. [arXiv:2308.12992 \[astro-ph.CO\]](#)
- [10] * H. Liu, **W. Qin**, G. W. Ridgway, and T. R. Slatyer. “Exotic energy injection in the early Universe. II. CMB spectral distortions and constraints on light dark matter”. *Phys. Rev. D*, 108(4):043531, 2023. [arXiv:2303.07370 \[astro-ph.CO\]](#)
- [9] * H. Liu, **W. Qin**, G. W. Ridgway, and T. R. Slatyer. “Exotic energy injection in the early Universe. I. A novel treatment for low-energy electrons and photons”. *Phys. Rev. D*, 108(4):043530, 2023. [arXiv:2303.07366 \[astro-ph.CO\]](#)
- [8] **W. Qin**, S. R. Geller, S. Balaji, E. McDonough, and D. I. Kaiser. “Planck constraints and gravitational wave forecasts for primordial black hole dark matter seeded by multifield inflation”. *Phys. Rev. D*, 108(4):043508, 2023. [arXiv:2303.02168 \[astro-ph.CO\]](#)
- [7] **W. Qin**, K. Schutz, A. Smith, E. Garaldi, R. Kannan, T. R. Slatyer, and M. Vogelsberger. “An Effective bias expansion for 21-cm cosmology in redshift space”. *Phys. Rev. D*, 106(12):123506, 2022. [arXiv:2205.06270 \[astro-ph.CO\]](#)
- [6] S. R. Geller, **W. Qin**, E. McDonough, and D. I. Kaiser. “Primordial black holes from multifield inflation with nonminimal couplings”. *Phys. Rev. D*, 106(6):063535, 2022. [arXiv:2205.04471 \[hep-th\]](#)
- [5] * H. Liu, **W. Qin**, G. W. Ridgway, and T. R. Slatyer. “Lyman- α constraints on cosmic heating from dark matter annihilation and decay”. *Phys. Rev. D*, 104(4):043514, August 2021. [arXiv:2008.01084 \[astro-ph.CO\]](#)
- [4] **W. Qin**, K. K. Boddy, and M. Kamionkowski. “Subluminal stochastic gravitational waves in pulsar-timing arrays and astrometry”. *Phys. Rev. D*, 103(2):024045, January 2021. [arXiv:2007.11009 \[gr-qc\]](#)
- [3] **W. Qin**, K. K. Boddy, M. Kamionkowski, and L. Dai. “Pulsar-timing arrays, astrometry, and gravitational waves”. *Phys. Rev. D*, D99(6):063002, March 2019. [arXiv:1810.02369 \[astro-ph.CO\]](#)
- [2] **W. Qin**, D. M. Nataf, N. Zakamska, P. R. Wood, and L. Casagrande. “The Mira-based distance to the Galactic centre”. *The Astrophysical Journal*, 865(1):47, August 2018. [arXiv:1808.01294 \[astro-ph.SR\]](#)
- [1] CMS Collaboration (including **W. Qin**). “Search for a new scalar resonance decaying to a pair of Z bosons in proton-proton collisions at $\sqrt{s}=13$ TeV”. *Journal of High Energy Physics*, 6:127, June 2018. [arXiv:1804.01939 \[hep-ex\]](#)

INVITED SEMINARS

- 2026, Apr 2 · Queen’s University McDonald Institute Seminar, “Not-quite-primordial black holes”
- 2026, Feb 27 · UIUC HEP Seminar, “Not-quite-primordial black holes”
- 2026, Feb 2 · University of Oklahoma Theory Seminar, “Not-quite-primordial black holes”
- 2026, Jan 26 · Stanford KIPAC Seminar, “Not-quite-primordial black holes”
- 2026, Jan 21 · LBNL Particle Seminar, “Not-quite-primordial black holes”
- 2025, Nov 5 · Cornell HEP Theory Seminar, “Not-quite-primordial black holes”
- 2025, Oct 14 · University of Valencia IFIC Seminar, “Not-quite-primordial black holes”
- 2025, Oct 9 · University of Barcelona ICCUB Seminar, “Not-quite-primordial black holes”
- 2025, Sep 16 · University of Delaware Seminar, “Not-quite-primordial black holes”
- 2025, Jun 16 · University of Amsterdam GRAPPA Colloquium,
“Using cosmic dawn to look beyond Lambda-CDM”

- 2025, May 1 · WashU Theory Seminar,
"CMB limits on decaying dark matter: going beyond the ionization threshold"
- 2025, Feb 27 · Stony Brook YITP Seminar,
"CMB limits on decaying dark matter: going beyond the ionization threshold"
- 2025, Jan 30 · Winnipeg Particle Theory Seminar,
"CMB limits on decaying dark matter: going beyond the ionization threshold"
- 2025, Jan 27 · University of Pennsylvania High Energy Theory Seminar,
"CMB limits on decaying dark matter: going beyond the ionization threshold"
- 2024, Dec 10 · Laboratoire Univers et Particules de Montpellier Seminar,
"CMB limits on decaying dark matter: going beyond the ionization threshold"
- 2024, Oct 31 · University of Pittsburgh HEP Seminar,
"CMB limits on decaying dark matter: going beyond the ionization threshold"
- 2024, Oct 1 · Johns Hopkins Theory Seminar,
"CMB limits on decaying dark matter: going beyond the ionization threshold"
- 2024, Apr 1 · Fermilab CPC Seminar,
"Probing exotic energy injection with the CMB and early star formation"
- 2024, Mar 5 · MIT/Tufts Cosmology Seminar,
"Probing exotic energy injection with the CMB and early star formation"
- 2024, Feb 20 · Ohio State University CCAPP Seminar,
"Probing exotic energy injection with the CMB and early star formation"
- 2023, Dec 4 · University of Toronto High-Energy Physics Seminar,
"Probing exotic energy injection with the CMB and early star formation"
- 2023, Nov 28 · Perimeter Institute Cosmology Seminar,
"Probing exotic energy injection with the CMB and early star formation"
- 2023, Nov 8 · SLAC Elementary Particle Physics Theory Seminar,
"Probing exotic energy injection with the CMB and early star formation"
- 2023, Nov 6 · UC Berkeley/LBNL 4D Seminar,
"Probing exotic energy injection with the CMB and early star formation"
- 2023, Oct 5 · Brown Theoretical Physics Center Seminar,
"Probing exotic energy injection with the CMB and early star formation"
- 2023, Oct 3 · Princeton Dark Cosmos Seminar,
"Probing exotic energy injection with the CMB and early star formation"
- 2023, Jul 20 · IPPP Durham Seminar,
"Probing exotic energy injection with the CMB (power) spectrum and early star formation"
- 2023, Jul 17 · Jodrell Bank Center for Astrophysics Special Cosmology Seminar,
"Probing exotic energy injection with the CMB (power) spectrum and early star formation"
- 2023, Apr 18 · Harvard & Smithsonian Center for Astrophysics Seminar,
"Cosmological probes of dark matter energy injection"

CONFERENCE PRESENTATIONS

- 2025, May 18-22 · EuCAPT Symposium, invited plenary talk, "Not-quite-primordial black holes"
- 2025, Jul 25 · Cosmology Summer Fest, "Not-quite-primordial black holes"
- 2024, Aug 25 · TeVPA, parallel talk,
"Birth of the first stars amidst decaying and annihilating dark matter"
- 2024, Mar 13 · Black Holes and Cosmology, contributed talk,

"Planck Constraints and Gravitational Wave Forecasts for Primordial Black Hole Dark Matter Seeded by Multifield Inflation"

- 2023, May 30 · Dark Matter, parallel talk,
"Probing exotic energy injection with CMB spectral distortions and early star formation"
- 2023, May 23 · PLANCK, parallel talk,
"Probing exotic energy injection with CMB spectral distortions and early star formation"
- 2023, May 9 · Pheno Symposium, parallel talk,
"An Effective Field Theory of 21 cm Radiation with Redshift Space Distortions"
- 2022, Aug 8 · TeVPA, parallel talk,
"An Effective Field Theory of 21 cm Radiation with Redshift Space Distortions"
- 2022, Jul 7 · ICHEP, parallel talk,
"Post-Recombination Spectral Distortions from Dark Matter Energy Injection"
- 2022, May 23 · EuCAPT Symposium, lightning talk,
"An Effective Field Theory of 21 cm Radiation with Redshift Space Distortions"
- 2022, May 10 · Pheno Symposium, parallel talk,
"Post-Recombination Spectral Distortions from Dark Matter Energy Injection"
- 2020, Aug 25 · Cosmology From Home, virtual parallel talk,
"Lyman-alpha constraints on Cosmic Heating from Dark Matter Annihilation and Decay"

INTERNAL TALKS

- 2025, May 5 · New York University CCPP Brown Bag,
"A new way to form supermassive black holes"
- 2025, Apr 1 · Simons Society of Fellows Retreat,
"Looking for Dark Matter: The World is your Calorimeter"
- 2025, Feb 18 · Institute for Advanced Study Amplitudes Group Meeting,
"Dark Matter Energy Injection: The World is your Calorimeter"
- 2024, Oct 21 · New York University CCPP Brown Bag,
"Dark Matter Energy Injection: The World is your Calorimeter"
- 2023, Oct 9 · Harvard Cosmology Journal Club,
"Probing exotic energy injection with the CMB and early star formation"
- 2023, Jul 13 · Cambridge University Cosmic Dawn group meeting,
"Probing exotic energy injection with the CMB (power) spectrum and early star formation"
- 2023, Apr 14 · MIT Center for Theoretical Physics, Dark Matter Journal Club,
"Cosmological probes of dark matter energy injection"
- 2023, Apr 10 · MIT Kavli Institute, Graduate Lunch Seminar,
"Cosmological probes of dark matter energy injection"
- 2022, Nov 15 · Technion Theory Group Meeting,
"Extending the Effective Field Theory of 21cm radiation"
- 2022, Oct 21 · MIT Center for Theoretical Physics, Graduate Lunch Seminar,
"Dark Matter at Cosmic Dawn: Probing BSM Physics with the 21 cm line"
- 2022, May 24 · Masui Group Meeting,
"Extending the Effective Field Theory of 21cm radiation"
- 2022, Feb 8 · HERA Theory Group Telecon,
"Extending the Effective Field Theory of 21cm radiation"
- 2021, Sep 28 · Harvard University, Particle Theory Journal Club,
"Introduction to 21cm Cosmology"

- 2021, May 14 · MIT Center for Theoretical Physics, BSM Journal Club,
"Redshift space distortions in an effective field theory of 21cm radiation"
- 2021, May 05 · MIT Kavli Institute, Graduate Lunch Seminar,
"Redshift space distortions in an effective field theory of 21cm radiation"
- 2020, Aug 14 · MIT Center for Theoretical Physics, BSM Journal Club,
"Lyman-alpha constraints on Cosmic Heating from Dark Matter Annihilation and Decay"
- 2019, Dec 04 · MIT Kavli Institute, Graduate Lunch Seminar,
"Pulsar timing arrays, astrometry, and gravitational waves"

WORKSHOPS

- 2025, Jun 15-19 · New Horizons in Primordial Black Hole Physics (CERN) — Invited Speaker
- 2025, Dec 8 - 12 · High Energy Theory in the Swamp (University of Florida) — Invited Speaker
- 2025, Sep 29 - Oct 3 · UNDARK Workshop (Instituto de Astrofisica de Canarias) — Invited Speaker
- 2025, Sep 17 - 19 · Primordial Black Holes: Theory Meets Experiment (University of Massachusetts Amherst, ACFI) — Speaker
- 2025, Jun 2 - 13 · Navigating New Horizons: Defining the Next Era of Particle Physics (Munich Institute for Astro- Particle- and BioPhysics) — Speaker
- 2025, Mar 3 - 6 · Particles vs. New Probes (Flatiron Institute)
- 2024, Nov 1 · Recent scientific developments in cosmology and particle physics (University of Pennsylvania)
- 2024, Feb 26 - 28 · Dark Matter, First Light (Perimeter Institute) — Invited Speaker

PROFESSIONAL ACTIVITIES AND SCHOOLS

- 2022 - present · Journal Referee: Journal of Cosmology and Astroparticle Physics, Journal of High Energy Physics, Physical Review D, Physical Review Letters
- 2022, Jun 6 - Jul 1 · TASI: Ten Years After the Higgs Discovery, Particle Physics Now and Future
- 2021, Jul 25 - Aug 20 · Les Houches Summer School: Dark Matter

UNDERGRADUATE MENTORSHIP/RESEARCH SUPERVISION

- 2023-2024 · Olivia Rosenstein, Massachusetts Institute of Technology
 (→ Fulbright Scholar at ENS Paris-Saclay and Ph.D. at Caltech)
- 2020-2024 · Clara Xu, Massachusetts Institute of Technology (→ Ph.D. at UC Berkeley)
- 2019-2022 · Eve Schoen, Massachusetts Institute of Technology (→ Ph.D. at UC Berkeley)

HIGH SCHOOL MENTORSHIP/RESEARCH SUPERVISION

- 2025 · Teddy Golden, The Spence School

COMMUNITY AND OUTREACH

- 2026 · Volunteer at the GO Project in New York City
- 2025 · Speaker at the New York City Cosmic Happy Hour
 - Ran the "Physics of Cooking" workshop for the APS Conference for Undergraduate Women

- and Gender Minorities in Physics (CU*IP) at New York University
- Member of local organizing committee for the APS Conference for Undergraduate Women and Gender Minorities in Physics (CU*IP) at New York University
- 2024 · Animation consultant for PBS NOVA documentary series
- 2023 · Co-chair of MIT Graduates Advising Graduate Admissions
- 2022 · Co-organizer of the MIT Dark Matter Journal Club and Coffee Hour
- Graduate Student Representative to MIT Graduate Admissions Committee (since 2020)
- Mentor for MIT UWIP-GWIP mentorship program (since 2019)
- 2021 · Mentor for MIT Physics Directed Reading Program
- Vice President of advocacy for the MIT Physics Graduate Student Council
- 2020 · Treasurer of the MIT Physics Graduate Student Council
- Graduate Women in Physics Representative to MIT Department Colloquium Committee
- Member of MIT Graduates Advising Graduate Admissions
- 2019 · Organizer for the MIT Center for Theoretical Physics Friday Lunch Seminar
- Member of the Johns Hopkins team for the 2019 International Olympiad in Theoretical Physics; placed 1st overall
- 2018 · Founder of the Johns Hopkins Physics Undergraduate Journal Club
- Member of the Johns Hopkins team for the 2018 International Olympiad in Theoretical Physics; placed 9th overall

TEACHING

- 2022 · Teaching Assistant at MIT (Modern Astrophysics 8.284)
- 2016-2018 · Teaching Assistant at Johns Hopkins University (General Physics 171.102 and 171.108)

IN THE NEWS

- 2026, Jan 23 · Jonathan O’Callaghan, *Quanta Magazine*
“Monster Neutrino Could Be a Messenger of Ancient Black Holes”
- 2026, Jan 23 · Adam Mann, *Live Science*
“ ‘A real revolution’: The James Webb telescope is upending our understanding of the biggest, oldest black holes in the universe”
- 2025, Sep 12 · Charlie Wood, *Quanta Magazine*
“A Single, ‘Naked’ Black Hole Rewrites the History of the Universe”

January 31, 2026